SHORT-PERIOD COMET

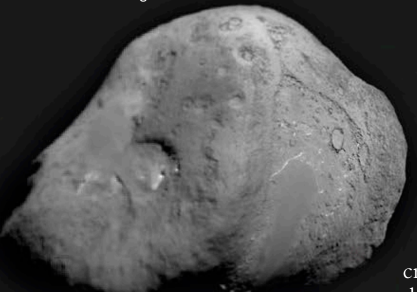
Tempel 1

CLOSEST APPROACH TO THE SUN 140 million miles (226 million km)

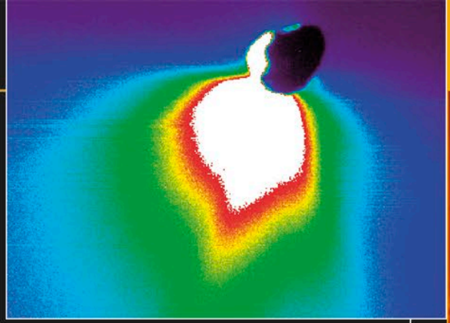
ORBITAL PERIOD 5.52 years

FIRST RECORDED April 3, 1867

This comet was first discovered in 1867 by the German astronomer Wilhelm Tempel, but after two reappearances, it disappeared because its orbit had been changed by close approaches to Jupiter. Following calculations by British astronomer Brian Marsden in 1963, the comet was rediscovered, and it has been followed ever since as it orbits between Mars and Jupiter.



TEMPLE NUCLEUS
The potato-shaped nucleus of Comet Tempel 1 was photographed by the Deep Impact probe in July 2005. The impactor hit between the craters at center right.



DEEP IMPACT
A fountain of dust, shown in false color, sprays off the nucleus of Comet Tempel 1. This image was taken on July 4, 2005, about 50 minutes after the comet's nucleus was hit by the impactor released by Deep Impact.

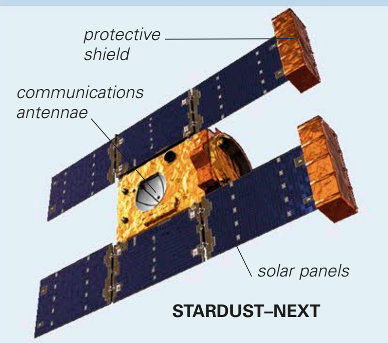
To find out what lies beneath the dusty crust of a comet's nucleus, NASA launched an ambitious mission to Tempel 1 in 2005. Called the Deep Impact probe, its aim was to punch a crater in the crust and uncover the subsurface ice, which is thought to have survived unchanged since the formation of the solar system. As the probe approached the nucleus of Tempel 1 in July 2005, it released a 820-lb (370-kg) copper impactor into its path. This collided with the nucleus at a speed of over 22,800 mph (36,000 kph), spraying out a fountain of dust and gas. Because the dust was so fine, about the same size as the particles in talcum powder, it appeared very bright—the comet temporarily brightened tenfold as a result, but it was still not visible to the naked eye. Deep Impact observed the ejected material to determine its composition. Most of the gas was steam (water vapor) and carbon dioxide at an initial temperature of over 1,340°F (720°C), resulting from the heat of the impact. There was so much dust that the crater formed by the impactor was hidden from view.

As Deep Impact flew past the comet, it took detailed images of the nucleus, which turned out to be shaped like a potato. It measured about 4¾ miles (7.5 km) long and 3 miles (5 km) across and rotated every 41 hours. It was very different from the nuclei of other comets that have been seen close up, such as Wild 2 and Borrelly. Surface features were visible, including a plateau fringed by a 66-ft (20-m) cliff (possibly the result of a landslide), and two apparent impact craters, each about 1,000 ft (300 m) wide. The impactor hit the nucleus between these craters.

Stardust, the craft that had collected dust samples from Comet Wild 2 (see p.218), was later sent to photograph the crater produced by Deep Impact. Renamed Stardust-NExT (for New Exploration of Tempel 1), it arrived in 2011, but saw little. It seems that the scar had been covered by dust that fell back onto it.

EXPLORING SPACE

STUDYING COMETS



Space-probe exploration of comets began at the last return of Halley's Comet, in 1986. Since then, probes have brought back samples of dust (Stardust-NExT, left) and hit the nucleus (Deep Impact). The next step was to orbit and land on a comet's nucleus—the mission of the European probe Rosetta. Comet probes are fitted with shields to protect them from the fast-moving specks of dust from the comets.

SHORT-PERIOD COMET

Hartley 2

CLOSEST APPROACH TO THE SUN 98 million miles (158 million km)

ORBITAL PERIOD 6.47 years

FIRST RECORDED March 15, 1986

This comet was discovered by British astronomer Malcolm Hartley while he was working at the Schmidt Telescope Unit at Siding Spring Observatory, Australia, in 1986. The nucleus of Comet Hartley 2 has been observed up close in a flyby from the Deep Impact probe.

Following its encounter with Comet Tempel 1 (see above), the Deep Impact probe was sent to take a closer look at Hartley 2. After a journey of 5 years, it arrived near the comet in November 2010 and flew past it at a distance of just under 435 miles (700 km). The spacecraft was not carrying a second impactor, so it could not hit the nucleus. Instead, research concentrated on the comet's appearance and composition.

Deep Impact's observations revealed that Hartley 2's nucleus, at only about 1.2 miles (2 km) long, was the smallest ever visited by a space probe. The comet is peanut-shaped, with two lobes that are connected by a smoother neck only about ¼ mile (0.4 km) wide. Jets of carbon-dioxide gas shoot out from the two lobes at either end of the nucleus, while water vapor is released from the middle. The levels of gas production also vary as the nucleus rotates over a period of about 18 hours. In addition, for the first time with any comet, the nucleus was seen to be shedding lumps of ice that ranged in size from golf balls to basketballs. Investigation by the Deep Impact probe also revealed larger blocks of ice up to 260 ft (80 m) high on the lobes of nucleus.

Although the spacecraft retained the name Deep Impact for this encounter, its extended space mission has been renamed EPOXI. This name comes from a combination of two acronyms: EPOCh, which stands for Extrasolar Planet Observation and Characterization, since its instruments observed a number of stars for evidence of transits by orbiting planets; and DIXI, short for Deep Impact Extended Investigation.

SPECTACULAR JETS
Huge jets of gas and dust spew from the elongated nucleus of Comet Hartley 2 as seen in this image from the Deep Impact probe. The image was taken on November 4, 2010, when the spacecraft was at it closest to the comet.

LONG-PERIOD COMET

Lovejoy

CLOSEST APPROACH TO THE SUN 515,000 miles (829,000 km)

ORBITAL PERIOD 565 years

FIRST RECORDED November 27, 2011

This sungrazer comet was discovered by an Australian amateur astronomer, Terry Lovejoy, less than 3 weeks before its closet approach to the Sun. Sungrazers are comets that skim so close to the Sun, they either evaporate in the intense heat or they crash into its surface. They are usually seen only by telescopes onboard satellites that monitor the region around the Sun, such as the Solar and Heliospheric Observatory (SOHO, see p.105).

In December 2011, Comet Lovejoy not only defied predictions by surviving passing so close to the Sun but emerged to become a brilliant object that could be seen from Earth. On December 16, 2011, satellites including the Solar Dynamics Observatory (SDO) watched as the comet passed the Sun at a distance of just over 82,000 miles (130,000 km). Over the following days, observers in the Southern Hemisphere were astounded as the comet moved away from the Sun, becoming visible in the morning skies, and grew a long, featherlike tail. Astronauts on the International Space Station got a particularly good view. Sungrazer comets are thought to be fragments of a much larger comet that broke up long ago, possibly in the 12th century. The pieces have continued to orbit the Sun, disintegrating further as they do so. They are also known Kreutz sungrazers, because they were first studied in the 19th century by German astronomer Heinrich Kreutz. Comet Lovejoy has a calculated orbital period of 565 years.

SPACE STATION VIEW
This view from the International Space Station shows Comet Lovejoy's tail extending upward from the horizon. The bands beneath are part of Earth's atmosphere.

SURVIVING THE SUN
Comet Lovejoy (circled) emerges from its passage through the Sun's inner corona as seen by NASA's Solar Dynamics Observatory.